

Calculus I Practice Final

A balanced twenty-five-question cumulative final with point values, study map, and complete solutions.

Revision: 2026-07-23 **Canonical page:** <https://bettergrades.net/subjects/math/calculus/practice-exams/calculus-1-final/>

Name: _____ **Date:** _____

1. Evaluate $\lim_{x \rightarrow 2} (x^2 + 3x - 1)$.

2. Evaluate $\lim_{x \rightarrow 4} (x^2 - 16)/(x - 4)$.

3. Choose k so $f(x) = kx + 1$ for $x < 2$ and $f(x) = 7$ for $x \geq 2$ is continuous.

4. Use the limit definition to find the derivative of x squared.

5. Differentiate $4x^5 - 3x^2 + 7$.

6. Differentiate $x^2 \sin x$.

7. Differentiate $(x + 1)/(x - 1)$.

8. Differentiate $(3x^2 + 1)^4$.

9. For $x^2 + y^2 = 25$, find dy/dx .

10. Find the tangent line to $y = x^3$ at $x = 2$.

11. Find c guaranteed by MVT for $f(x) = x^2$ on $[1, 3]$.

12. Find critical numbers of $x^3 - 3x$.

13. Where is $x^3 - 3x$ increasing?

14. Find the inflection point of $x^3 - 6x^2$.
15. A rectangle has perimeter 40. Find maximum area.
16. A circle radius grows at 2 cm/s. Find dA/dt at $r=5$.
17. Use linearization at 9 to approximate $\sqrt{9.2}$.
18. Find $\int (6x^2 - 4)dx$.
19. Evaluate $\int_0^2 3x^2 dx$.
20. Differentiate $F(x) = \int_1^x \cos(t^2)dt$.

21. Evaluate $\int 2x(x^2 + 1)^3 dx$.
22. Find the area under $y=x$ on $[0,3]$.
23. Find the average value of x squared on $[0,3]$.
24. If $v(t)=3t$ squared- $6t$, find displacement from 0 to 3.
25. Explain why a differentiable function is continuous.